

Nibble Security: Node.js Connect CSRF bypass abusing methodOverride middleware

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Content

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In the [previous post](#), I discussed the importance of well-written documentation and uncomplicated APIs suggesting that poor documentation and negligence should be considered as silent threats.

Almost a year ago, I reported the following issue to the Node.js Connect's maintainers. To me, this is a perfect example of the risks of an incomplete API documentation that doesn't clearly warn the user of potential side-effects. Please note that in the recent releases of Express, *connect-csrf* is now called *csrf* *and* **methodOverride* is now *method-override*. Different names, same API.

Disclosure timeline

This issue was reported to Senchalabs on 07/25/2013. Despite my requests to add a warning in the online documentation, there's still no indication of potential side-effects in [Connect MethodOverride](#). On 09/07/2013, this advisory was also published by the [NodeSecurity](#) community.

Unfortunately, I don't think that the issue raised the adequate level of attention as suggested by the many vulnerable applications that I've encountered.

Technical details

Connect's *methodOverride* middleware allows an HTTP request to override the HTTP verb with the value of the `_method` post parameter or with the `**x-http-method-override**` header. As the

declaration order of middlewares determines the execution stack in Connect, it is possible to abuse this functionality in order to bypass the standard Connect's anti-CSRF protection.

Considering the following code:

```
...  
app.use express.csrf()  
...  
app.use express.methodOverride()
```

Connect's CSRF middleware does not check CSRF tokens in case of idempotent verbs (GET/HEAD/OPTIONS, see [csurf/index.js](#)). As a result, it is possible to bypass the security control by sending a GET request with a POST MethodOverride header or parameter.

```
GET / HTTP/1.1  
[..]  
_method=POST
```

The workaround is clearly to disable methodOverride or make sure that it takes precedence over other middleware declarations.

Adam Baldwin made an [eslint plugin](#) that you can use to identify this issue.

Update 06/04: Douglas W. pointed out that it's probably a good idea to move to method-override version 2+ (<https://www.npmjs.org/package/method-override#readme>). The documentation has been updated with a reference to this issue.

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